

- Description

This module allows teachers to record final grades for students, ensuring that:

1. The final grade **must be between 0 and 100**.
2. If a valid grade is entered, **the system records the grade**.
3. If an invalid grade is entered, **the grade is rejected**.

- Specification

module StudentFinalGradeManagement

var grade: int

process recordFinalGrade(grade: int)

functional scenario form:

grade >= 0 and grade <= 100 && Valid grade entry, final grade recorded

||

grade < 0 && Invalid grade, grade rejected

||

grade > 100 && Invalid grade, grade rejected

end process

end module

- Java Code

```
package code;
public class c04_StudentFinalGradeManagement {
    public static boolean recordFinalGrade(int grade) {
        if (grade >= 0 && grade <= 100) {
            System.out.printf("Final grade recorded successfully! Grade
: %d%n", grade);
            return true;
        } else {
            System.out.println("Grade recording failed! Invalid grade."
);
            return false;
        }
    }
    public static void main(String[] args) {
        // testcase-VT:
        recordFinalGrade(72); // grade >= 0 && grade <= 100
        recordFinalGrade(-2); // grade < 0
        recordFinalGrade(223); // grade > 100

        // testcase-FT:
        // recordFinalGrade(4156828);
        // recordFinalGrade(2);
        // recordFinalGrade(126);
```

```

        // testcase-Z3:
//      recordFinalGrade(4);
//      recordFinalGrade(-1);
//      recordFinalGrade(101);

        // testcase-UVT:
//      recordFinalGrade(41);
//      recordFinalGrade(-2);
//      recordFinalGrade(126);
    }
}

```

- Coverage report

1. Test case number: 3
2. Vibration testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	812	816
✓ src	0.5 %	4	812	816
✓ code	0.5 %	4	812	816
✓ c04_StudentFinalGradeManagement.jl	100.0 %	4	0	4
✓ c04_StudentFinalGradeManagement.jl	100.0 %	4	0	4
main(String[])	100.0 %	0	0	0
recordFinalGrade(int)	100.0 %	4	0	4

3. V-Method testing branch coverage: 91.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.2 %	33	15,477	15,510
✓ src	0.2 %	33	15,477	15,510
✓ code	0.2 %	33	15,477	15,510
✓ c04_StudentFinalGradeManagement.jl	91.7 %	33	3	36
✓ c04_StudentFinalGradeManagement.jl	91.7 %	33	3	36
main(String[])	100.0 %	10	0	10
recordFinalGrade(int)	100.0 %	23	0	23

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.4 %	3	813	816
✓ src	0.4 %	3	813	816
✓ code	0.4 %	3	813	816
✓ c04_StudentFinalGradeManagement.jl	75.0 %	3	1	4
✓ c04_StudentFinalGradeManagement.jl	75.0 %	3	1	4
recordFinalGrade(int)	75.0 %	3	1	4
main(String[])	100.0 %	0	0	0

5. Fuzz testing instruction coverage: 91.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.2 %	33	15,477	15,510
✓ src	0.2 %	33	15,477	15,510
✓ code	0.2 %	33	15,477	15,510
✓ c04_StudentFinalGradeManagement.jl	91.7 %	33	3	36
✓ c04_StudentFinalGradeManagement.jl	91.7 %	33	3	36
main(String[])	100.0 %	10	0	10
recordFinalGrade(int)	100.0 %	23	0	23

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c04_StudentFinalGradeManagement.java	100.0 %	4	0	4
▼ c04_StudentFinalGradeManagement	100.0 %	4	0	4
recordFinalGrade(int)	100.0 %	4	0	4
main(String[])		0	0	0

7. Z3 instruction coverage: 91.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.1 %	33	22,053	22,086
▼ src	0.1 %	33	22,053	22,086
▼ code	0.1 %	33	22,053	22,086
▼ c04_StudentFinalGradeManagement.java	91.7 %	33	3	36
▼ c04_StudentFinalGradeManagement	91.7 %	33	3	36
main(String[])	100.0 %	10	0	10
recordFinalGrade(int)	100.0 %	23	0	23

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c04_StudentFinalGradeManagement.java	100.0 %	4	0	4
▼ c04_StudentFinalGradeManagement	100.0 %	4	0	4
recordFinalGrade(int)	100.0 %	4	0	4
main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 91.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.1 %	33	22,053	22,086
▼ src	0.1 %	33	22,053	22,086
▼ code	0.1 %	33	22,053	22,086
▼ c04_StudentFinalGradeManagement.java	91.7 %	33	3	36
▼ c04_StudentFinalGradeManagement	91.7 %	33	3	36
main(String[])	100.0 %	10	0	10
recordFinalGrade(int)	100.0 %	23	0	23